

# Adaptive Fraud Detection with Dynamic Thresholding and Cost-Sensitive Learning

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**Abstract** — We present a complete supervised ML pipeline for credit card fraud detection on the Kaggle Credit Card Fraud dataset (284,807 transactions; 0.17% fraud prevalence). Three classifiers are trained and compared: Logistic Regression (LR), Random Forest (RF), and Histogram-based Gradient Boosted Trees (GBT). Three class-imbalance strategies are evaluated: balanced class weighting, SMOTE, and random under-sampling. We introduce dynamic threshold tuning on the validation set and a cost-sensitive evaluation framework (FN = 500 vs. FP = 10). The best model—RF with balanced class weights at a tuned threshold of 0.192—achieves **PR-AUC = 0.8116**, **Recall = 82.4%**, and **Precision = 84.7%** on the held-out test set.

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## 1. Introduction

Credit card fraud causes billions of euros in annual losses. Automated detection reduces this, but the problem presents two fundamental challenges. First, the extreme class imbalance (577:1 in this dataset) renders standard accuracy uninformative: labelling every transaction as legitimate achieves 99.83% accuracy while catching zero fraud. Second, the two error types carry asymmetric costs: a missed fraud (false negative, FN) costs far more than a false alarm (false positive, FP) that triggers a manual review.

This work addresses both challenges through appropriate primary metrics (PR-AUC and ROC-AUC), three imbalance-handling strategies, and a cost-sensitive threshold selection procedure that minimises expected business cost on the validation set.

## 2. Dataset and Preprocessing

### 2.1. Dataset

The Kaggle Credit Card Fraud Detection dataset [1] contains 284,807 European cardholder transactions (September 2013), with 492 fraud cases (0.173%). Features V1–V28 are anonymised PCA components (zero-mean, decorrelated); Time and Amount are in original units. No missing values; 1,081 duplicate rows retained.

### 2.2. Preprocessing Pipeline

The following strict ordering prevents data leakage:

1. **Split first.** Stratified 70/15/15 train/validation/test split (199,356 / 42,729 / 42,722 samples), preserving 0.173% fraud in each subset.
2. **Scale on train only.** StandardScaler fitted on Time and Amount from training data; applied to all splits. Features V1–V28 require no re-scaling (PCA output).
3. **Resample train only.** SMOTE and under-sampling applied within the training split only, never touching validation or test.

## 3. Methodology

### 3.1. Models

**Logistic Regression (LR).** Linear baseline optimising cross-entropy with L2 regularisation ( $C = 1.0$ ). `class_weight='balanced'` reweights each fraud sample by  $\approx 577\times$  relative to legitimate samples.

**Random Forest (RF).** Ensemble of 100 trees trained on bootstrap samples with random feature subsets. Captures nonlinear feature interactions; same balanced class weighting.

**Gradient Boosted Trees (GBT).** `HistGradientBoostingClassifier` (sklearn, 200 iterations). Additively builds trees to correct ensemble residuals. Two variants: balanced class weighting (GBT-bal) and SMOTE-resampled training data (GBT-SMOTE).

### 3.2. Imbalance Handling

Three strategies were compared on LR to isolate resampling effects: (1) `class_weight='balanced'`; (2) SMOTE [2], which interpolates between minority samples to achieve 1:1 training ratio; and (3) random under-sampling, which discards majority samples until 1:1.

### 3.3. Threshold Tuning and Cost-Sensitive Evaluation

For threshold  $t$ , predicted class is  $\hat{y} = \mathbf{1}[\hat{p} \geq t]$ . We sweep  $t \in [0.01, 0.99]$  (200 steps) on the validation set and report:

- **F1-optimal threshold:**  $t_{F1}^* = \arg \max_t F1(t)$
- **Cost-optimal threshold:**  $t_{cost}^* = \arg \min_t [FN(t) \cdot C_{FN} + FP(t) \cdot C_{FP}]$

with  $C_{FN} = 500$  (average fraud loss) and  $C_{FP} = 10$  (manual review cost per transaction). All threshold selection is done on the validation set; the test set is evaluated exactly once.

## 4. Results

### 4.1. Resampling Strategy Comparison

Table ?? shows LR performance under the three imbalance strategies at the default threshold of 0.50. Class weighting and SMOTE achieve similar PR-AUC; random under-sampling underperforms due to the large information loss from discarding 199,012 majority-class training samples (99.8% of legitimate transactions). SMOTE is therefore applied to GBT as the secondary variant.

**Table 1:** Resampling strategy comparison on LR (validation set, threshold=0.50).

| Strategy       | PR-AUC | Recall | Fraud F1 |
|----------------|--------|--------|----------|
| Class weight   | 0.7520 | 0.9189 | 0.1265   |
| SMOTE          | 0.7599 | 0.9324 | 0.1209   |
| Under-sampling | 0.6986 | 0.9324 | 0.0865   |

### 4.2. Model Comparison at F1-Optimal Threshold

Table ?? compares all model variants at their respective F1-optimal thresholds on the validation set. RF achieves the highest PR-AUC (0.8637), confirming that nonlinear feature interactions are important for fraud detection. GBT-SMOTE achieves the highest F1 (0.8794) but at a lower PR-AUC (0.8533). LR lags substantially on both PR-AUC and F1.

The F1-optimal thresholds vary widely: RF finds its optimum at 0.192 (well below the default 0.5), while LR requires 0.990. This divergence reflects differences in probability calibration across model families.

**Table 2:** All models at their F1-optimal thresholds (validation set). Best values in bold.

| Model            | PR-AUC        | ROC-AUC       | Thr.         | F1            | Recall        | Prec.         |
|------------------|---------------|---------------|--------------|---------------|---------------|---------------|
| LR (bal.)        | 0.7520        | 0.9860        | 0.990        | 0.7273        | 0.8649        | 0.6275        |
| <b>RF (bal.)</b> | <b>0.8637</b> | 0.9514        | <b>0.192</b> | 0.8767        | <b>0.8649</b> | 0.8889        |
| GBT (bal.)       | 0.7840        | <b>0.9822</b> | 0.951        | 0.8531        | 0.8243        | 0.8841        |
| GBT (SMOTE)      | 0.8533        | 0.9729        | 0.960        | <b>0.8794</b> | 0.8378        | <b>0.9254</b> |

### 4.3. Threshold Tuning and Cost Analysis

Figure 1 shows F1, Precision, and Recall as a function of threshold for all models. Figure 2 shows total business cost vs. threshold. Because  $C_{FN} \gg C_{FP}$  (50:1), the cost-optimal thresholds are substantially lower than the F1-optimal thresholds across all models—the system prefers flagging more transactions (higher FP) to avoid missing fraud (lower FN).

For RF, the cost-optimal threshold drops to 0.044, catching one additional fraud (FN: 13  $\rightarrow$  12) while increasing FP from 11 to 48, for a net saving of 130 (6,480 vs. 6,610). Table ?? summarises the comparison.

**Table 3:** F1-optimal vs. cost-optimal threshold analysis for all models on validation set.

| Model          | F1-opt. Thr. | Cost-opt. Thr. |
|----------------|--------------|----------------|
| LR (balanced)  | 0.990        | 0.985          |
| RF (balanced)  | 0.192        | 0.044          |
| GBT (balanced) | 0.951        | 0.837          |
| GBT (SMOTE)    | 0.960        | 0.251          |

### 4.4. Final Test Set Evaluation

RF (balanced) was selected as the best model by validation PR-AUC=0.8637. Table ?? reports test-set performance at both thresholds.

**Table 4:** RF (balanced) on the held-out test set (74 fraud; 42,648 legitimate).

| Threshold         | ROC-AUC | PR-AUC | F1     | Recall | Prec.  | Cost () |
|-------------------|---------|--------|--------|--------|--------|---------|
| 0.192 (F1-opt.)   | 0.9376  | 0.8116 | 0.8356 | 0.8243 | 0.8472 | 6,610   |
| 0.044 (cost-opt.) | 0.9376  | 0.8116 | 0.6739 | 0.8378 | 0.5636 | 6,480   |

At the F1-optimal threshold the model catches 61 of 74 frauds (82.4% recall) with only 11 false alarms. Figure 3 shows the confusion matrix, ROC curve, and Precision-Recall curve on the test set.

## 5. Error Analysis

The 13 false negatives (missed frauds at threshold 0.192) had a mean model score of 0.021, vs. 0.826 for detected frauds. Feature-distribution analysis of top discriminative features (V14, V17, V12) shows that missed frauds have values close to the legitimate distribution, indicating they are genuinely ambiguous cases rather than model failures. Their transaction amounts were slightly lower than detected frauds, consistent with small test charges being harder to distinguish from low-value legitimate transactions. The 11 false positives had a mean score of 0.44, just above the decision threshold, suggesting they are legitimate transactions with unusual PCA projections resembling known fraud signatures.

## 6. Discussion and Conclusions

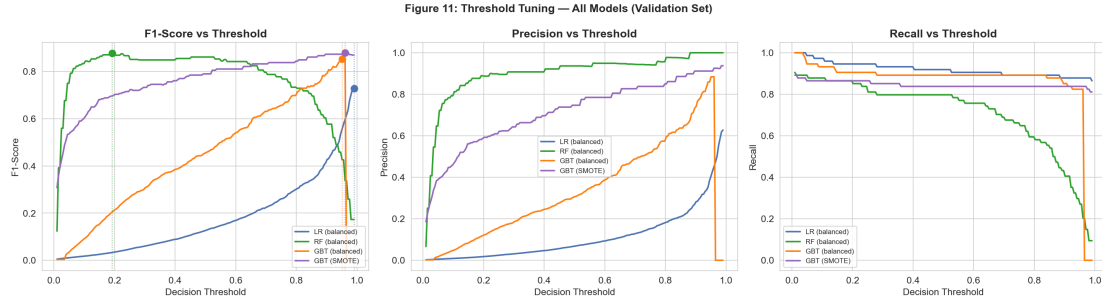
**Nonlinear models are essential.** RF and GBT outperform LR by 0.11 PR-AUC, confirming that fraud signals arise from nonlinear feature combinations that a linear boundary cannot capture.

**Threshold tuning is as important as model choice.** Dynamic tuning of the RF threshold from 0.50 to 0.192 improves F1 from 0.8613 to 0.8767 on the validation set with no model change.

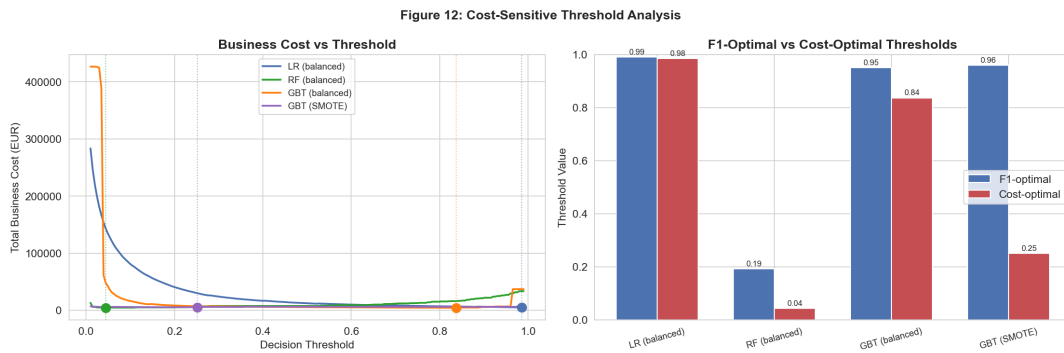
**Cost-sensitive analysis changes the operating point.** The 50:1 FN/FP cost ratio shifts the RF optimal threshold from 0.192 to 0.044. Whether this trade-off is acceptable depends on the institution's capacity for manual fraud review.

**SMOTE vs. class weighting.** Both achieve similar PR-AUC on LR (+0.008 for SMOTE). On GBT, class weighting yields higher PR-AUC while SMOTE yields higher F1; class weighting appears better calibrated across all thresholds.

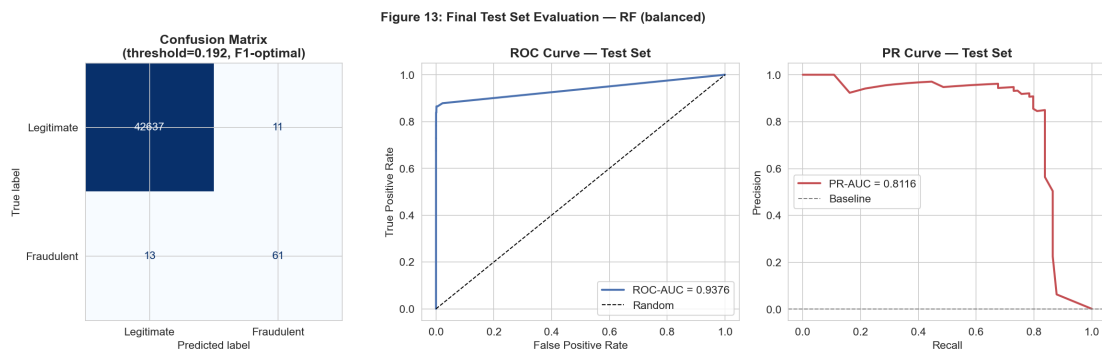
**Limitations and future work.** Features V1–V28 are opaque PCA components; future work with raw transaction features would enable actionable model explanations. Hyperparameter tuning (grid search over tree depth, learning rate) and time-aware cross-validation (respecting temporal ordering) remain as extensions. Online learning [4] could address distribution shift as fraud strategies evolve.



**Figure 1:** F1-score, Precision, and Recall vs. decision threshold for all four models on the validation set. Dots mark the F1-maximising threshold per model.



**Figure 2:** Left: Total business cost vs. threshold for all models (validation set). Right: Comparison of F1-optimal vs. cost-optimal thresholds. Cost-optimal thresholds are consistently lower due to the 50:1 FN/FP cost ratio.



**Figure 3:** Final test set evaluation for RF (balanced) at threshold 0.192. Confusion matrix (61 TP, 11 FP, 13 FN, 42,637 TN), ROC curve (AUC = 0.9376), and PR curve (AUC = 0.8116).

## References

- [1] ULB Machine Learning Group, *Credit Card Fraud Detection Dataset*, Kaggle, 2018. <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

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- [4] A. Dal Pozzolo, O. Caelen, R.A. Johnson, G. Bontempi, “Calibrating Probability with Undersampling for Unbalanced Classification,” *IEEE SSCI*, 2015.